Research Proposal: A Probabilistic Digital Twin for ICU Vital-Sign Forecasting

Distribution-aware deterioration risk via conditional diffusion models

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Proposed period: November 2026 – July 2027 (volunteer research assistants) **Status:** Initial draft for discussion

1. Summary

Deterministic forecasters (LSTM/GRU/Transformer) collapse volatile patient futures into a single, clinically uninformative average. We propose to train a conditional diffusion model that generates many plausible future trajectories of ICU vital signs, and to score deterioration risk from the tails of that distribution (e.g., the probability that mean arterial pressure falls below 60 mmHg). We aim to show that this distribution-aware risk score is better calibrated and gives earlier warning than point forecasts or direct classifiers, validated across two ICU cohorts, and to submit the work for publication.

2. Where the Project Stands

Our Fall course project (CSE 455/555) produced a working prototype on the PhysioNet 2019 Sepsis Challenge data (5 vitals: HR, MAP, SpO2, SBP, Resp; hourly, 48-hour windows): a CSDI-style conditional DDPM with a 1D-convolutional denoiser that samples 50 future trajectories per patient, an attention-based LSTM baseline, and exploratory sepsis vs. non-sepsis analysis. Our review identified the following gaps between the prototype and a publishable study:

Gap	Why it matters
No probabilistic metrics	Fan charts are not evidence; CRPS, calibration, coverage and sharpness are needed for measurable claims.
No clinical risk / alarm module	The core idea (converting sampled futures into early warnings) has not yet been built or evaluated.
Patient-level data leakage	Windows were split randomly, so the same patient can appear in both train and test sets.
Basic preprocessing	Fixed-divisor normalization and zero-filling ignore informative missingness in ICU data.
Limited baselines and rigor	Only an LSTM baseline; no seeds, variance, confidence intervals or ablations.
Single cohort	No external validation beyond PhysioNet 2019 training set A.

3. Proposed Work (initial, subject to revision)

Research question: Does a tail-risk score derived from diffusion-generated trajectories detect deterioration earlier and more reliably than (i) a deterministic forecast with the same thresholds, (ii) a direct discriminative classifier, and (iii) clinical early-warning scores (NEWS/MEWS)?

- **Leak-free data pipeline:** patient-level splits, train-only normalization, missingness masks, explicit cohort and exclusion criteria.
- **Probabilistic evaluation harness:** CRPS, energy score, quantile loss, interval coverage, reliability diagrams, sharpness; per horizon and per vital.
- **Clinical risk module:** threshold-crossing probabilities and time-to-threshold aggregated into a risk score; evaluated by AUROC/AUPRC, lead time and decision-curve analysis.
- **Stronger models and baselines:** CSDI-style time × variable attention denoiser; baselines including CSDI, TimeGrad, deep ensembles, MC-dropout and quantile regression.
- **External validation:** PhysioNet 2019 set B (a separate hospital system) immediately; MIMIC-IV as the headline external cohort, subject to credentialed access.

Intended contribution: a distributional risk-scoring framework, a calibration and early-warning evaluation, and cross-cohort validation. Architectural novelty is optional; candidate extensions include fast sampling (DDIM) and longer forecast horizons.

4. Key References

- Tashiro et al., *CSDI: Conditional Score-based Diffusion Models for Probabilistic Time Series Imputation*, NeurIPS 2021 (arXiv:2107.03502). **Foundation of our Fall prototype.**

- Chang et al., *A Transformer-based Diffusion Probabilistic Model for Heart Rate and Blood Pressure Forecasting in ICU*, CMPB 2024 (arXiv:2301.06625). **Closest prior work; we will benchmark against it and differentiate through risk framing, calibration and the sepsis cohort.**
- Supporting: Rasul et al., *TimeGrad*, ICML 2021; Alcaraz & Strodthoff, *SSSD*, TMLR 2023. Further literature will be added as the work progresses.

5. Roles and Responsibilities

Each member will contribute about 20 hours per week. Roles are stable, but we will pair-program, review each other's work and rotate meeting updates.

Member	Workstream	Responsibilities	Start
Nandini	Data, cohorts & evaluation infrastructure	Leak-free pipeline; set B and MIMIC-IV preparation and credentialing; evaluation harness and baseline runner shared by all experiments.	Nov 2026
Anjaneya	Models & diffusion research	Denoiser upgrades (attention/transformer/SSM); baseline implementations; training infrastructure, seeds, ablations, fast sampling.	On EAD approval
Faham	Clinical risk module, experiments & writing	Risk-scoring framework, clinical labels and lead-time analysis; results and figures; literature review; leads paper and poster drafting.	On EAD approval

Anjaneya and Faham are awaiting their EADs. Until then they will do uncounted preparatory reading and setup, and Nandini will build the shared foundation they will plug into.

6. Proposed Timeline

Months	Focus	Milestone
Nov	Reproduce prototype, repository cleanup, literature review, dataset decision	Clean repository; reading complete
Nov – Dec	Leak-free pipeline, evaluation harness, baselines	Reproducible benchmark table
Jan – Feb	Model upgrades; risk-scoring module v1; first end-to-end results	Working risk score with metrics
Mar – Apr	External validation (set B, MIMIC-IV), ablations, calibration study	Workshop paper submission (e.g., ML4H)
May – Jun	Complete results; full paper; internal review with Prof. Nwogu	Submission-ready draft
Jul	Polish and submit (target: CHIL, JBHI or CMPB); poster	Paper submitted

7. Support Requested and Logistics

- **MIMIC-IV access:** serving as our supervisor/reference for PhysioNet credentialing (we will complete CITI training).
- **Compute:** we plan to use our own GPU or Colab Pro+ if needed.
- **Guidance:** regular check-ins at your preferred cadence, and feedback on scope and target venues.
- **Hours tracking:** a shared Trello board with a weekly log per member (date, hours, task, deliverable), visible to you.